Ch 10.3: Convolutional Neural Nets

Lecture 31 - CMSE 381

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

Fri, Apr 10, 2026

Announcements

Last time:

- Multilayer NN
- pyTorch

This lecture:

- CNNs

Final countdown:

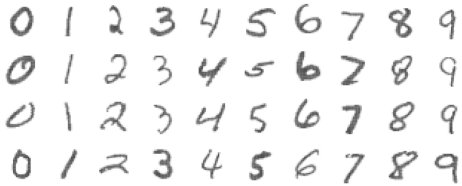
- HW #6 was due Sunday 4/12
- Exam 3 is 4/20
- Project is due 4/24

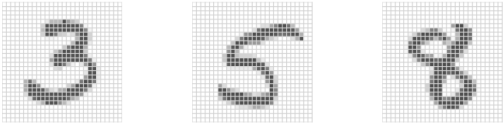
21	W	3/18	Polynomial & Step Functions	7.1-7.2		
22	F	3/20	Step Functions; Basis functions; Start Splines	7.2-7.4		
23	M	3/23	Regression Splines	7.4		Q7
24	W	3/25	Decision Trees	8.1		
25	F	3/27	Random Forests	8.2.1, 8.2.2	HW #5 Due Sun 3/29	
26	M	3/30	Maximal Margin Classifier	9.1		Q8
27	W	4/1	SVC	9.2		
28	F	4/3	SVM	9.3, 9.4		
29	M	4/6	Single Layer NN	10.1		Q9
30	W	4/8	Multi Layer NN	10.2		
31	F	4/10	CNN	10.3	HW #6 Due Sun 4/12	
32	M	4/13	Unsupervised learning / clustering	12.1, 12.4		Q10
33	W	4/15	Virtual: Project Office Hours			
	F	4/17	***Review***			
	M	4/20	**Midterm #3**			
	W	4/22				
	F	4/24			**Project Due**	

Section 1

Last time: Neural Nets

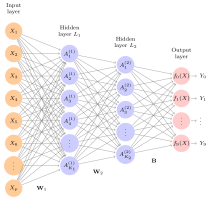
MNIST





- Goal: Build a model to classify images into their correct digit class
- Each image has $p = 28 \cdot 28 = 784$ pixels
- Each pixel is grayscale value in [0,255]
- Data converted into column order
- Output represented by one-hot vector $Y = (Y_0, Y_1, \cdots, Y_9)$
- 60K training images, 10K test images

Neural network architecture for MNIST

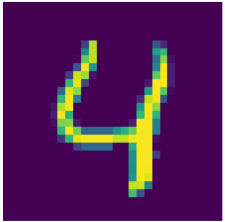


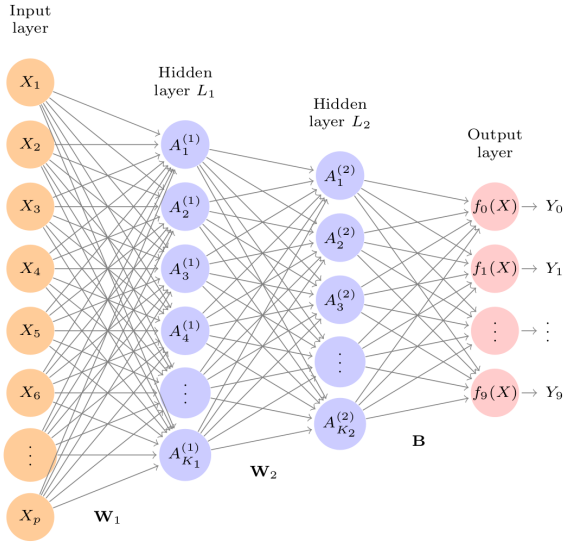
$$
\begin{aligned}
A_k^{(1)} &= h_k^{(1)}(X) \\
&= g(w_{k0}^{(1)} + \textstyle\sum_{j=1}^{p} w_{kj}^{(1)} X_j) \\
A_\ell^{(2)} &= h_\ell^{(2)}(X) \\
&= g(w_{\ell 0}^{(2)} + \textstyle\sum_{k=1}^{K_1} w_{\ell k}^{(2)} A_k^{(1)}) \\
Z_m &= \beta_{m0} + \textstyle\sum_{\ell=1}^{K_2} \beta_{m\ell} h_\ell^{(2)}(X) \\
&= \beta_{m0} + \textstyle\sum_{\ell=1}^{K_2} \beta_{m\ell} A_\ell^{(2)},
\end{aligned}
$$

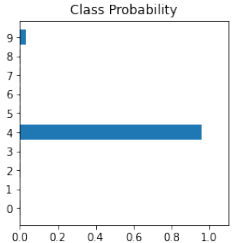
$$
f_m(X) = \Pr(Y = m|X) = \frac{e^{Z_m}}{\sum_{\ell=0}^{9} e^{Z_\ell}},
$$

- Two hidden layers.
- Softmax for classification output
- We used L_1 has 128 units; L_2 has 64
- 10 output variables due to class labeling
- Result is we are training approx 110K weights

MNIST learning







What will you learn today?

- What is the architecture of a convolutional neural network?
  - You should be able to describe the types of layers, how they are arranged, and the purpose of each of them.
- How does convolution work?
  - You should be able to compute the result of the convolution between a simple matrix and a simple filter by hand.
- What information does Convolutional Neural Network capture through convolution?

Section 2

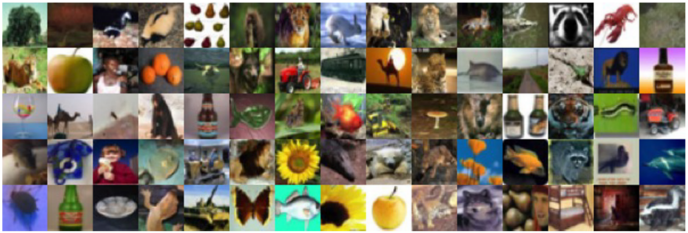
Convolutional Neural Network

Last time: Flattening the image – why is it not great?

$$\begin{pmatrix} 1 & 1 & 0 \\ 4 & 2 & 1 \\ 0 & 2 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 \\ 1 \\ 0 \\ 4 \\ 2 \\ 1 \\ 0 \\ 2 \\ 1 \end{pmatrix}$$

- Just ignore the fact that we have a picture
- Loses a lot of the structure that we might want for more complicated
- Answer is CNNs, which take in image data and build a neural net that specifically tracks that structure

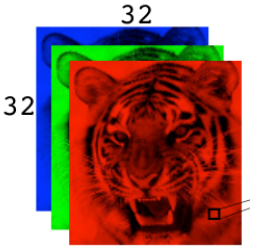
Example data set: CIFAR100 Data



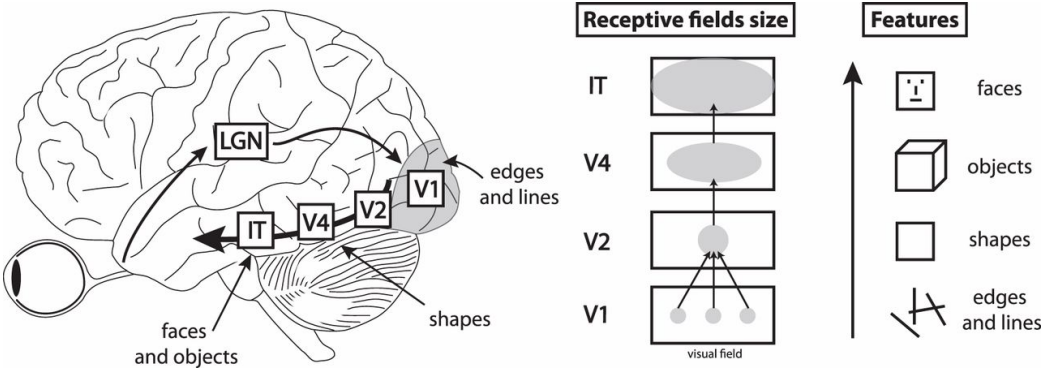
- 60,000 images: 50K training, 10K test
- Labels with 20 super classes (e.g. aquatic mammals)
- 5 classes per super class (beaver, dolphin, otter, seal, whale)
- Images are 32x32

Image channel data

- 3 eight bit numbers per pixel (red, green, blue)
- Numbers are organized in a *feature map* , which is 3 dimensional array
- First two directions are spatial (32x32)
- Third is the *channel* axis representing teh three colors



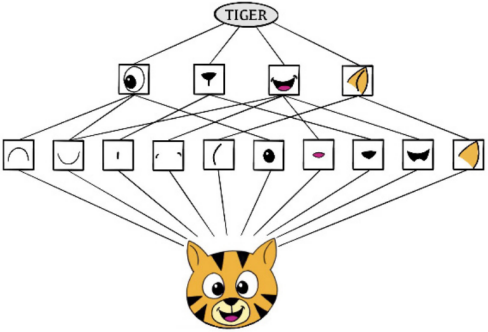
How does your brain do it? The visual hierarchy



Mauro Manassi, Bilge Sayim, Michael H. Herzog; When crowding of crowding leads to uncrowding. Journal of Vision 2013;13(13):10. https://doi.org/10.1167/13.13.10.

CNNs

- really similar to the human visual hierarchy
- earlier layers Identify low level features on the input image such as small edges, patches of color, etc
- Low level features combined into higher-level features (parts of ears, eyes, etc) in later layers
- Does this using special types of hidden layers: *convolution* layers and *pooling layers*



Convolution layer

Convolution Filter

Original Image:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \\ j & k & l \end{bmatrix}$$

Convolution filter:

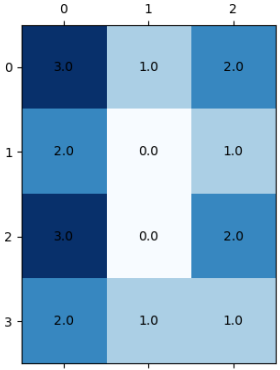
$$\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$$

Convolved Image

$$\begin{bmatrix} a\alpha + b\beta + d\gamma + e\delta & b\alpha + c\beta + e\gamma + f\delta \\ d\alpha + e\beta + g\gamma + h\delta & e\alpha + f\beta + h\gamma + i\delta \\ g\alpha + h\beta + j\gamma + k\delta & h\alpha + i\beta + k\gamma + l\delta \end{bmatrix}$$

Note we are just doing inner product here, which measure(ish) similar

Convolution Filter Example



Filter:

$$\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

- $\begin{bmatrix} 1 & 3 \\ 0 & 3 \\ 1 & 3 \end{bmatrix}$
- Convolved image highlights regions that resemble the convolution filter.
- Conv or filtering is traditional signal processing, but in CNN the filter is learned (mentioned again later)
- Note that the result is smaller than original

Same example, different filter

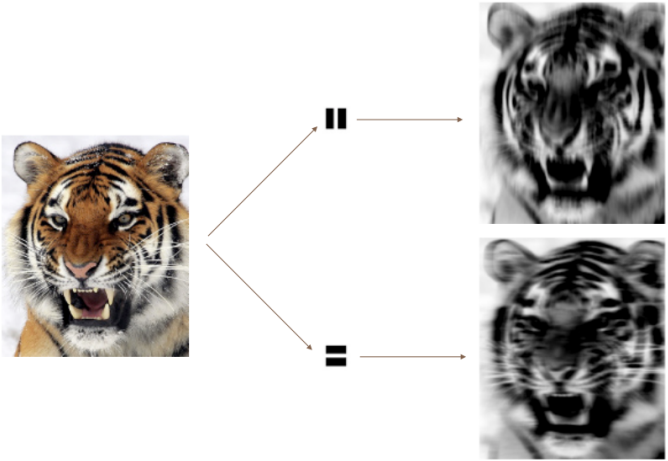
What is the convolved image?

Filter:

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

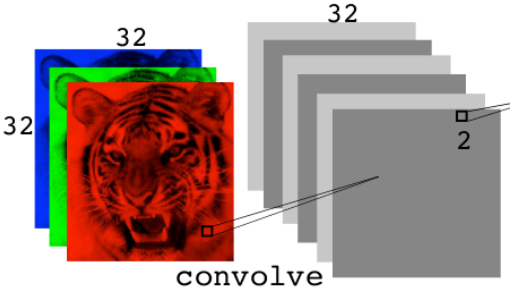
- $\begin{bmatrix} 4 & 3 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}$

Convolution filter: Bigger example



- 192x179 image of tiger
- Convolution filters are 15x15 images with mostly zero (black) narrow strip of ones (white)
- Highlights areas of the tiger that resemble the filter

Convolution layer



- Image is in color, so 3 channels
- This means the convolution filter will have three channels, one per color, each of dimension 3x3
- Results of the three convolutions are summed to form a 2-dimensional output. Color is "forgotton" at this point
- K different filters here means K 2-d output maps, which can then be treated as channels
- Padding to make the updated image the same size

More notes on convolution

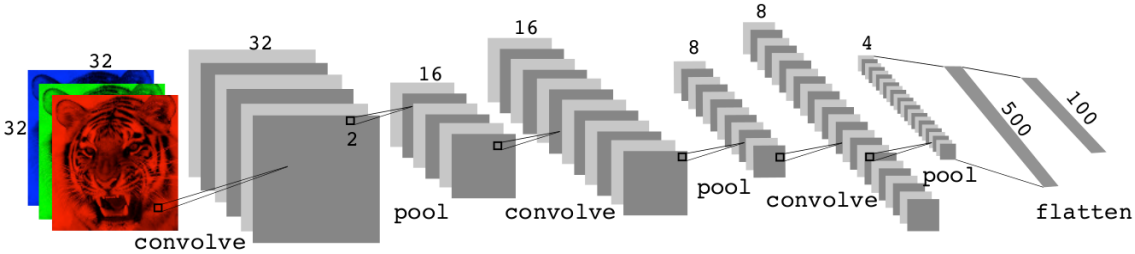
- Filters idea from image processing
- Novelty in DL is that the choice of filters are learned
- Filter weights: one hidden unit per pixel in convolved image; but more constrained than that
- same weight used for all patches for the images and channels of only m-by-m weights per filter, where m is patch size.
- Applying ReLU to convolved image. Sometimes viewed as separate layer in CNN, and then called a detector layer.

Pooling layers

- Condense a large image into a smaller summary image
- Lots of options, but max pooling is common:
- For each 2x2 non-overlapping block of pixels, get a single new pixel with the maximum value from the block
- Provides location invariance: so long as large value in one of the four pixels, the whole block registers as a large value

$$\text{Max pool} \begin{bmatrix} 1 & 2 & 5 & 3 \\ 3 & 0 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 1 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$$

Putting it together to make a CNN



- Start with three channels
- get a new channel at first hidden layer for each filter. Padding to keep the same size
- Next is max pool layer,
- convolve & pool several more times

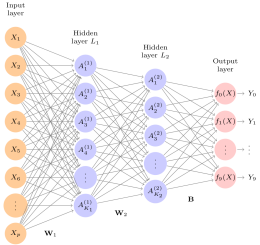
- Since size of image decreased, we increase number of filters in the next convolve layer to compensate
- Flattened, then fed into 1 or more fully-connected layers
- Last is softmax activation for the 100 classes

https://poloclub.github.io/cnn-explainer/

Coding

TL;DR

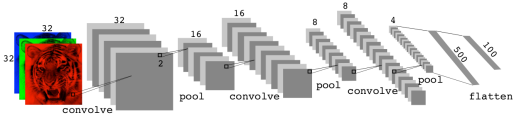
Feed Forward Neural Net



$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j),$$

- Combines input data using learned weights
- Linear combo of those to get output
- Sometimes softmax to get probability of classification

CNN



- Specialized NN
- Gets next layer via
  - Convolution layer
  - Pooling Layer
  - Fully connected layer

Next time

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